Safety Data Sheet
acc. to OSHA HCS

Printing date 08/21/2026 *Version number 1* *Reviewed on 08/21/2026*

1 Identification

. ***Product identifier***

. ***Trade name: GRINOVA H XE 12311 BLUE L2435.1***

. ***Article number:*** 106916
. ***Application of the substance / the mixture*** Polymeric material in granular form for thermoplastic processing

. ***Details of the supplier of the safety data sheet***
. ***Manufacturer/Supplier:***
EMS-CHEMIE AG CH-7013 Domat/Ems
Tel.: +41 81 632 78 88 Fax: +41 81 632 76 65
E-Mail: welcome@emsgrivory.com

. ***Information department:***
Abteilung Umweltschutz und Sicherheit
Tel.: +41 81 632 78 88 Fax: +41 81 632 76 65
E-Mail: produktsicherheit@emsservices.ch
. ***Emergency telephone number:***
EMS-CHEMIE AG CH-7013 Domat/Ems
Phone: +41 81 632 78 88 Fax: +41 81 632 76 65
Poison Center Switzerland Zürich Phone: +41 44 251 51 51

2 Hazard(s) identification

. ***Classification of the substance or mixture***
The product is not classified, according to the Globally Harmonized System (GHS).

. ***Label elements***
. ***GHS label elements*** Void
. ***Hazard pictograms*** Void
. ***Signal word*** Void
. ***Hazard statements*** Void
. ***Classification system:***
. ***NFPA ratings (scale 0 - 4)***



Health = 0
Fire = 1
Reactivity = 0

. ***HMIS-ratings (scale 0 - 4)***



Health = 0
Fire = 1
Reactivity = 0

. ***Other hazards***
. ***Results of PBT and vPvB assessment***
. ***PBT:*** Not applicable
. ***vPvB:*** Not applicable

3 Composition/information on ingredients

. ***Chemical characterization: Mixtures***
. ***Description:***
Polyamide
Additives / Modifiers

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. ***Dangerous components:*** Void

4 First-aid measures

. ***Description of first aid measures***
. ***General information:*** No special measures required.
. ***After inhalation:*** Supply fresh air; consult doctor in case of complaints.
. ***After skin contact:***
If skin irritation continues, consult a doctor.
After contact with the molten product, cool rapidly with cold water.
Do not pull solidified product away from the skin.
Seek medical treatment.
. ***After eye contact:***
Rinse opened eye for several minutes under running water. If symptoms persist, consult a doctor.
. ***After swallowing:*** If symptoms persist consult doctor.
. ***Information for doctor:***
. ***Most important symptoms and effects, both acute and delayed*** No further relevant information available.
. ***Indication of any immediate medical attention and special treatment needed***
No further relevant information available.

5 Fire-fighting measures

. ***Extinguishing media***
. ***Suitable extinguishing agents:***
CO2, extinguishing powder or water spray. Fight larger fires with water spray or alcohol resistant foam.
. ***For safety reasons unsuitable extinguishing agents:*** none
. ***Special hazards arising from the substance or mixture***
In case of fire, the following can be released:
Carbon monoxide (CO)
Nitrogen oxides (NOx)
Carbon dioxide (CO2)
Hydrogen cyanide (HCN)
Phosporous compounds
In certain fire conditions, traces of other toxic gases cannot be excluded.
. ***Advice for firefighters***
. ***Protective equipment:*** Use self-contained breathing apparatus and protective fire fighting clothing.
. ***Additional information***
Dispose of fire debris and contaminated fire fighting water in accordance with official regulations.

6 Accidental release measures

. ***Personal precautions, protective equipment and emergency procedures***
Particular danger of slipping on leaked/spilled product.
. ***Environmental precautions:***
Do not allow to enter sewers/ surface or ground water.
Do not allow to penetrate the ground/soil.
. ***Methods and material for containment and cleaning up:***
Pick up mechanically.
Dispose of the collected material according to regulations.
. ***Reference to other sections***
See Section 7 for information on safe handling.
See Section 8 for information on personal protection equipment.
See Section 13 for disposal information.

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7 Handling and storage

. ***Handling:***
. ***Precautions for safe handling***
Store in cool, dry place in tightly closed receptacles.
Ensure good ventilation/exhaustion at the workplace.
Extractors are required on all machines used for thermal processing or splinter removal processes.
Prevent formation of dust.
Any deposit of dust which cannot be avoided must be regularly removed.
. ***Information about protection against explosions and fires:*** Protect against electrostatic charges.

. ***Conditions for safe storage, including any incompatibilities***
. ***Storage:***
. ***Requirements to be met by storerooms and receptacles:*** No special requirements.
. ***Information about storage in one common storage facility:*** Not required
. ***Further information about storage conditions:*** Protect from humidity and water.
. ***Storage class:*** 10-13
. ***Specific end use(s)*** No further relevant information available.

8 Exposure controls/personal protection

. ***Additional information about design of technical systems:*** No further data; see section 7.

. ***Control parameters***
. ***Components with limit values that require monitoring at the workplace:***
National workplace exposure limits for airborne particles (respirable and inhalable dust) have to be observed.

. ***Exposure controls***
. ***Personal protective equipment:***
. ***General protective and hygienic measures:***
Wash hands before breaks and at the end of work.
Avoid skin contact with the liquefied material.
Do not inhale dust / smoke / mist.
Avoid close or long term contact with the skin.
. ***Breathing equipment:***
Not necessary if room is well-ventilated.
Use suitable respiratory protective device in case of insufficient ventilation.
Filter A/P2
. ***Protection of hands:***
Protective gloves with chemical protection level type A according EN ISO 374 and cut protection level A3/C according EN 388.
. ***For the permanent contact in work areas without heightened risk of injury (e.g. Laboratory) gloves made of the following material are suitable:***
nitrile rubber (≥ 1.7 mm)
. ***Eye protection:***
Safety glasses
EN 166
. ***Body protection:*** Protective work clothing

9 Physical and chemical properties

. ***Information on basic physical and chemical properties***
. ***General Information***
. ***Appearance:***

Form:	Granulate
Color:	According to product specification
. ***Odor:***	Nearly odorless

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. **Change in condition** **Melting point/Melting range:**	305-325 °C (581-617 °F)
. **Flash point:**	Not applicable
. **Decomposition temperature:**	> 350 °C (> 662 °F)
. **Ignition temperature:**	Product is not self igniting.
. **Danger of explosion:**	Product does not present an explosion hazard.
. **Density at 20 °C (68 °F):**	> 1.0 g/cm³ (> 8.35 lbs/gal)
. **Solubility in / Miscibility with Water:**	Insoluble.
. **Other information**	No further relevant information available.

10 Stability and reactivity

. **Reactivity** Reacts with strong acids and oxidizing agents.
. **Chemical stability** No decomposition if used and stored according to specifications.
. **Possibility of hazardous reactions**
As the product is supplied it is not capable of dust explosion; however, enrichment with fine dust causes risk of dust explosion.
. **Conditions to avoid** No further relevant information available.
. **Incompatible materials:**
Strong acids.
Strong oxidizing agents.
. **Hazardous decomposition products:** No hazardous decomposition products known.

11 Toxicological information

. **Information on toxicological effects**
. **Acute toxicity:**
. **Primary irritant effect:**
. **on the skin:** No irritant effect.
. **on the eye:** No irritant effect.
. **Sensitization:** No sensitizing effects known.
. **Additional toxicological information:**
When used and handled according to specifications, the product does not have any harmful effects according to our experience and the information provided to us.

. **Carcinogenic categories**

. **IARC (International Agency for Research on Cancer)**
None of the ingredients is listed.

. **NTP (National Toxicology Program)**
None of the ingredients is listed.

. **OSHA-Ca (Occupational Safety & Health Administration)**
None of the ingredients is listed.

12 Ecological information

. **Toxicity**
. **Aquatic toxicity:** No further relevant information available.
. **Persistence and degradability** No further relevant information available.

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. ***Behavior in environmental systems:***
. ***Bioaccumulative potential*** No further relevant information available.
. ***Mobility in soil*** No further relevant information available.
. ***Additional ecological information:***
. ***General notes:***
Do not allow undiluted product or large quantities of it to reach ground water, water course or sewage system.
. ***Results of PBT and vPvB assessment***
. ***PBT:*** Not applicable
. ***vPvB:*** Not applicable
. ***Other adverse effects*** No further relevant information available.

13 Disposal considerations

. ***Waste treatment methods***
. ***Recommendation:*** Disposal according to local statutory requirements.

. ***Uncleaned packagings:***
. ***Recommendation:*** Disposal must be made according to official regulations.

14 Transport information

. ***UN-Number***	
. ***DOT, ADR/RID, ADN, IMDG, IATA***	Void
. ***UN proper shipping name***	
. ***DOT, ADR/RID, ADN, IMDG, IATA***	Void
. ***Transport hazard class(es)***	
. ***DOT, ADR/RID, ADN, IMDG, IATA***	
. ***Class***	Void
. ***Packing group***	
. ***DOT, ADR/RID, IMDG, IATA***	Void
. ***Environmental hazards:***	
. ***Marine pollutant:***	No
. ***Special precautions for user***	Not applicable
. ***Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code***	Not applicable
. ***Transport/Additional information:***	Not dangerous according to ADR/RID/IMDG/IATA.
. ***UN "Model Regulation":***	Void

15 Regulatory information

. ***Safety, health and environmental regulations/legislation specific for the substance or mixture***
. ***Sara***

. ***Section 355 (extremely hazardous substances):***
None of the ingredient is listed.

. ***Section 313 (Specific toxic chemical listings):***

1314-98-3	zinc sulphide

. ***TSCA (Toxic Substances Control Act):***
All components have either the value ACTIVE or are exempt from TSCA inventory requirements.

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. **Hazardous Air Pollutants**

None of the ingredients is listed.

. **Proposition 65**

. **Chemicals known to cause cancer:**

None of the ingredients is listed.

. **Chemicals known to cause reproductive toxicity for females:**

None of the ingredients is listed.

. **Chemicals known to cause reproductive toxicity for males:**

None of the ingredients is listed.

. **Chemicals known to cause developmental toxicity:**

None of the ingredients is listed.

. **Cancerogenity categories**

. **EPA (Environmental Protection Agency)**

1314-98-3	zinc sulphide	D, I, II

. **NIOSH-Ca (National Institute for Occupational Safety and Health)**

None of the ingredients is listed.

. **GHS label elements** Void
. **Hazard pictograms** Void
. **Signal word** Void
. **Hazard statements** Void
. **Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

16 Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

. **Department issuing SDS:** Environment and safety department (U+S)
. **Contact:**
Abteilung Umweltschutz und Sicherheit
Tel.: +41 81 632 78 88 Fax: +41 81 632 76 65
E-Mail: produktsicherheit@emsservices.ch
. **Abbreviations and acronyms:**
RID: Règlement international concernant le transport des marchandises dangereuses par chemin de fer (Regulations Concerning the International Transport of Dangerous Goods by Rail)
ICAO: International Civil Aviation Organisation
ADR: Accord relatif au transport international des marchandises dangereuses par route (European Agreement Concerning the International Carriage of Dangerous Goods by Road)
IMDG: International Maritime Code for Dangerous Goods
DOT: US Department of Transportation
IATA: International Air Transport Association
EINECS: European Inventory of Existing Commercial Chemical Substances
ELINCS: European List of Notified Chemical Substances
CAS: Chemical Abstracts Service (division of the American Chemical Society)
NFPA: National Fire Protection Association (USA)
HMIS: Hazardous Materials Identification System (USA)
PBT: Persistent, Bioaccumulative and Toxic
vPvB: very Persistent and very Bioaccumulative
NIOSH: National Institute for Occupational Safety
OSHA: Occupational Safety & Health
TLV: Threshold Limit Value
PEL: Permissible Exposure Limit
REL: Recommended Exposure Limit